

Errata File for Second Edition

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This errata file contains a list of errors, short clarifications and modifications to the manuscript's second edition.

1. Page 7, first sentence below equation (1.5): "This means that there is a 95% probability that this interval contains the true parameter θ ".

This statement is technically incorrect for the following reason. A $100(1 - \alpha)\%$ confidence interval for θ is an interval defined by the lower and upper bounds $(l(Y), u(Y))$, where Y represents the data, a random variable, such that

$$\Pr[l(Y) < \theta < u(Y) | \theta] = 1 - \alpha,$$

where θ is the fixed, unknown parameter and α is the significance level of the test. The probability is computed with respect to the distribution of Y . Once the data are observed and $Y = y$, the expression

$$\Pr[l(y) < \theta < u(y) | \theta] = 1 - \alpha,$$

does not make sense since there are no random variables involved within the square brackets. It is therefore technically incorrect to assert that the observed interval $(l(y), u(y))$ contains θ with probability $(1 - \alpha)$. The interval $(l(y), u(y))$ either contains or does not contain θ . However, if $\alpha = 0.05$, one can be quite confident that the unobserved θ is within the observed $(l(y), u(y))$ and act accordingly.

The remaining statements in the paragraph are correct.

2. Page 170, bottom half of the page, bracket missing:
CHANGE:

$$\text{Var}(\hat{a} - a) = E_y(\text{Var}(a|y))$$

TO.

$$\text{Var}(\hat{a} - a) = E_y(\text{Var}(a|y))$$

3. Page 171, the first partial derivative below Eq. (4.62).

CHANGE:

$$\frac{\partial \ln p(a, b|y, b_0, a_0, B, G, R)}{\partial b} = -B^{-1}(b - b_0) + XR^{-1}(y - Xb - Za)$$

TO:

$$\frac{\partial \ln p(a, b|y, b_0, a_0, B, G, R)}{\partial b} = -B^{-1}(b - b_0) + X'R^{-1}(y - Xb - Za)$$

4. Page 172, top.

CHANGE: "When B^{-1} implying $B = \infty \dots$ "

TO: "When and if $B^{-1} = 0$ implying $B = \infty \dots$ "

5. Page 172,

CHANGE:

Finally, add and subtract $\hat{\theta}'(W'R^{-1}W + \Sigma)\hat{\theta}$ and rearrange to yield

$$y'R^{-1}y + (\theta - \hat{\theta})'(W'R^{-1}W + \Sigma)(\theta - \hat{\theta}) - \hat{\theta}'(W'R^{-1}W + \Sigma)\hat{\theta} + \theta'_0\Sigma\theta_0$$

TO:

Finally, add and subtract $\hat{\theta}'(W'R^{-1}W + \Sigma)\hat{\theta}$ and rearrange to yield

$$y'R^{-1}y + (\theta - \hat{\theta})'(W'R^{-1}W + \Sigma)(\theta - \hat{\theta}) - \hat{\theta}'(W'R^{-1}W + \Sigma)\hat{\theta} + \theta'_0\Sigma\theta_0$$

6. Page 172, equation at the bottom,

CHANGE:

$$\ln p(\theta|y, \theta_0, B, G, R) \propto -\frac{1}{2}(\theta - \hat{\theta})'(W'R^{-1}W + \Sigma)(\theta - \hat{\theta})'$$

TO:

$$\ln p(\theta|y, \theta_0, B, G, R) \propto -\frac{1}{2}(\theta - \hat{\theta})'(W'R^{-1}W + \Sigma)(\theta - \hat{\theta})$$

7. Page 263,

CHANGE,

$$v_\beta^* = v + 1,$$

TO:

$$v_\beta^* = v_\beta + 1,$$

8. Page 265,
Delete Subsection "An Alternative Parametrisation" (up to and including eqn (5.82)).

9. Page 353, Eq. (7.41), remove "comma" before " $i = 1, \dots, m$ "
CHANGE:

$$y|\mu, \alpha, \delta, \sigma^2 \sim N\left(1\mu + \sum_{i=1}^m X_i(\alpha_i \delta_i), I\sigma^2\right), \quad i = 1, \dots, m \quad (7.41)$$

TO:

$$y|\mu, \alpha, \delta, \sigma^2 \sim N\left(1\mu + \sum_{i=1}^m X_i(\alpha_i \delta_i), I\sigma^2\right), \quad i = 1, \dots, m \quad (7.41)$$

10. Page 361,
CHANGE:

"In terms of p -values, a threshold of 0.05 guarantees that the expected number of false positives over the whole family of m tests is less than or equal to $0.05m$, far too large for modern genomewide studies."

TO:

"In terms of p -values, a threshold of 0.05 guarantees that the expected number of false positives over the whole family of m tests is $0.05m$, far too large for modern genomewide studies."

11. Page 361,
CHANGE:

"Then for the whole family of m tests, when all the m hypotheses are true nulls, the expected number of false positives is less than or equal to $m(\alpha/m) = \alpha$."

TO:

"Then for the whole family of m tests, when all the m hypotheses are true nulls, the expected number of false positives is equal to $m(\alpha/m) = \alpha$."

12. Page 363,
CHANGE:

Then the expected number of false discoveries over the m tests is

$$E(V) = E\left(\sum_{i \in M_0} V_i\right) = \sum_{i \in M_0} E(V_i) \leq m_0 \alpha_B.$$

TO:

Then the expected number of false discoveries over the m tests is

$$E(V) = E\left(\sum_{i \in M_0} V_i\right) = \sum_{i \in M_0} E(V_i) = m_0 \alpha_B.$$

13. Page 373, paragraph above Eq. (8.23), defines the number of false discoveries $V(A)$, given that k hypotheses are rejected, as binomially distributed $Bi(k, \Pr(H = 0|z \in A))$. Another definition can be found on page 376, paragraph above Eq. (8.33), concerning the expected number of false discoveries. Both definitions may benefit from a little explanation that is provided below.

Background

Let the binary random variable $V_i = 1$ if hypothesis i is rejected and if hypothesis i is a true null; $V_i = 0$, otherwise. Using the notation on page 363, we can write $V_i = I(H_i = 0, R_i = 1)$, where $R_i = 1$ if $z_i \in A$ and z_i is the observed value of the scalar random variable Z_i . Treating the hypotheses as random, it follows that V_i is a Bernoulli random variable with probability $\Pr(H_i = 0, Z_i \in A)$; that is $V_i \sim Br(\Pr(H_i = 0, Z_i \in A))$.

$$V_i \sim Br(\Pr(H_i = 0, Z_i \in A)).$$

- **Page 373.** Consider the binary random variable $V_i|R_i = 1$ that takes the value 1 if $H_i = 0$ and 0 otherwise. Then $[V_i|R_i = 1] \sim Br(\Pr(H_i = 0|Z_i \in A))$, a Bernoulli distributed random variable. If in a series of independent trials the total number of rejections is $R = k$, among these k rejections, the total number of false discoveries is binomially distributed:

$$[V|R = k] \sim Bi(k, \Pr(H = 0|Z \in A)).$$

The expected number of false discoveries among the k rejections is

$$E(V|R = k) = k \Pr(H = 0|Z \in A).$$

The subscript i has been omitted because $\Pr(H_i = 0|Z_i \in A)$ is the same for each i .

- **Page 376.** Consider an experiment where m independent hypotheses are tested. Given the definition of V_i and the independence assumption, the total number of false discoveries, $V = \sum_{i=1}^m V_i$, is then binomially distributed

$$V \sim Bi(m, \Pr(H = 0, Z \in A)),$$

with expected value

$$E(V) = m \Pr(H = 0, Z \in A) = m \Pr(Z \in A|H = 0)\pi_0,$$

where $\pi_0 = \Pr(H = 0)$, the prior probability of a true null.

Note that: (i) hypotheses H_i are treated as random (with a common prior π_0), (ii) random vectors (H_i, Z_i) , $i = 1, 2, \dots, m$, are independently distributed, (iii) the rejection region A is data-independent and fixed in advance.